

Environmental Toxicology

THE EFFECT OF pH ON CHRONIC AQUATIC NICKEL TOXICITY IS DEPENDENT ON THE pH ITSELF: EXTENDING THE CHRONIC NICKEL BIOAVAILABILITY MODELS

CHARLOTTE NYS,^{*,†} COLIN R. JANSSEN,[†] PATRICK VAN SPRANG,[‡] and KAREL A.C. DE SCHAMPHELAERE[†]

[†]Faculty of Bioscience Engineering, GhentToxLab, Ghent University, Ghent, Belgium

[‡]Arche, Ghent, Belgium

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Abstract: The environmental quality standard for Ni in the European Commission's Water Framework Directive is bioavailability based. Although some of the available chronic Ni bioavailability models are validated only for $\text{pH} \leq 8.2$, a considerable fraction of European surface waters has a $\text{pH} > 8.2$. Therefore, the authors investigated the effect of a change in pH from 8.2 to 8.7 on chronic Ni toxicity in 3 invertebrate (*Daphnia magna*, *Lymnaea stagnalis*, and *Brachionus calyciflorus*) and 2 plant species (*Pseudokirchneriella subcapitata* and *Lemna minor*). Nickel toxicity was almost always significantly higher at pH 8.7 than at pH 8.2. To test whether the existing chronic Ni bioavailability models developed for $\text{pH} \leq 8.2$ can be used at higher pH levels, Ni toxicity at pH 8.7 was predicted based on Ni toxicity observed at pH 8.2. This resulted in a consistent underestimation of toxicity. The results suggest that the effect of pH on Ni^{2+} toxicity is dependent on the pH itself: the slope of the pH effect is steeper above than below pH 8.2 for species for which a species-specific bioavailability model exists. Therefore, the existing chronic Ni bioavailability models were modified to allow predictions of chronic Ni toxicity to invertebrates and plants in the pH range of 8.2 to 8.7 by applying a pH slope (S_{pH}) dependent on the pH of the target water. These modified Ni bioavailability models resulted in more accurate predictions of Ni toxicity to all 5 species (within 2-fold error), without the bias observed using the bioavailability models developed for $\text{pH} \leq 8.2$. The results of the present study can decrease the uncertainty in implementing the bioavailability-based environmental quality standard under the Water Framework Directive for high-pH regions in Europe. *Environ Toxicol Chem* 2016;35:1097–1106. © 2015 SETAC

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INTRODUCTION

Chronic toxicity of Ni is strongly dependent on the physicochemistry of freshwater environments [1–4]. More specifically, Mg and Ca are in general protective against chronic Ni toxicity to aquatic organisms, whereas chronic Ni toxicity increases at high pH. This knowledge recently resulted in validated chronic Ni bioavailability models for several aquatic organisms: *Daphnia magna* (invertebrate) [3], *Ceriodaphnia dubia* (invertebrate) [5], *Oncorhynchus mykiss* (fish) [2], and *Pseudokirchneriella subcapitata* (algae) [4]. These bioavailability models consider the protective effect of certain cations on Ni^{2+} toxicity (Ca^{2+} and Mg^{2+}), which compete with Ni^{2+} for binding at the biotic ligand and are in that respect similar to biotic ligand models [6,7]. Unlike traditional biotic ligand models, however, the effect of pH on Ni toxicity is generally not modeled as a competition effect between Ni^{2+} and H^+ for binding at the biotic ligand. Instead, the effect of pH on Ni^{2+} toxicity is superimposed on the Ca^{2+} and Mg^{2+} competition effects [2–5]. It has been shown that chronic Ni bioavailability models can be extrapolated to many other species, including several algal [8], fish [2], and cladoceran species [9], but also to *Lymnaea stagnalis* (snail), *Brachionus calyciflorus* (rotifer), *Chironomus tentans* (midge), and *Lemna minor* (aquatic plant) [10]. The chronic Ni bioavailability models are currently used in the European Union risk assessment procedures to determine bioavailability-normalized predicted no effect concentrations (PNECs) [11] and to determine compliance with the

“bioavailable” environmental quality standard of Ni in the European Commission's Water Framework Directive [12], which is dependent on the water chemistry of the considered water body [13,14]. However, these bioavailability models are validated only within certain water chemistry ranges. For instance, the *D. magna* and the *P. subcapitata* models are validated for a pH up to 8.0 and 8.2, respectively. Yet 28% and 15% of the European surface waters have a pH above 8.0 and 8.2, respectively [15]. This means that a substantial number of the European waters falls outside the bioavailability model applicability range. Hence, for these waters, the use of the bioavailability normalization approach to determine site-specific PNECs or environmental quality standards in regulatory frameworks involves scientific uncertainty. Moreover, the Water Framework Directive [16,17] requires that environmental quality standards must be protective of at least 95% of the European surface waters, therefore requesting the consideration of surface waters with $\text{pH} > 8.0$.

Currently, only the *O. mykiss* [2] and the *C. dubia* bioavailability models [5] are validated for pH values above 8.2. Although the *O. mykiss* model was shown to be applicable over the pH range 5.4 to 8.5 [2], this was not the case for the *C. dubia* model. The effect of pH on Ni^{2+} toxicity to *C. dubia* has been shown to be dependent on the pH itself; that is, the slope of the pH effect was higher between pH 8.2 and 8.7, compared with that observed below pH 8.2 [5]. Therefore, the *C. dubia* bioavailability model developed for pH below 8.2 cannot be extrapolated to confidently predict Ni toxicity above pH 8.2. However, using a different pH slope (Equation 1) for pH up to 8.2 ($S_{\text{pH} \leq 8.2} = 0.86$) and pH above 8.2 ($S_{\text{pH} > 8.2} = 1.82$) in the *C. dubia* model resulted in more accurate predictions of Ni toxicity over a pH range from pH 6.5 up to pH 8.7 [5]. There are no chronic toxicity data

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* Address correspondence to chnys.nys@ugent.be

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for any other species in this pH range that allow testing of whether the other chronic Ni bioavailability models (valid for up to pH 8.2) can be safely extrapolated up to pH 8.7 or whether a modification of the existing bioavailability models is needed.

The aim of the study was 2-fold. The first aim was to investigate the effect of a pH change from 8.2 (bioavailability model boundary) to pH 8.7 on the chronic Ni toxicity for species for which bioavailability models exist or for species for which cross-species validation of the chronic bioavailability models has been demonstrated [10]. Based on the current bioavailability models, we hypothesized that the toxicity of the Ni^{2+} ion will be higher at pH 8.7 compared with pH 8.2. The second aim was to test whether the current bioavailability models can be extrapolated to pH 8.7 or whether modifications of the current chronic bioavailability models are needed. Therefore, we tested the effect of a change in pH from 8.2 to 8.7 on chronic Ni toxicity to 5 species: a micro-alga (*P. subcapitata*), a cladoceran (*D. magna*), a higher plant (*L. minor*), a snail (*L. stagnalis*), and a rotifer (*B. calyciflorus*). Finally, we evaluated the performance of the existing chronic Ni bioavailability models for predicting Ni toxicity at pH 8.7 based on the observed Ni toxicity in the pH 8.2 treatment.

MATERIALS AND METHODS

General design

Chronic ecotoxicity experiments were performed for 5 species (exposure duration and endpoint in parentheses): *P. subcapitata* (72-h growth rate), *D. magna* (21-d reproduction), *L. minor* (7-d growth rate), *L. stagnalis* (28-d growth rate), and *B. calyciflorus* (48-h population growth rate). All experiments were conducted following standard protocols (except for *L. stagnalis*, for which no standard protocol is currently available). A species-specific experimental medium was used following the standard test protocols of the respective species. However, all test media were modified to contain equal concentrations of Ca and Mg (1.5 mM and 0.25 mM, respectively) for all test species. For every species, Ni toxicity was investigated at 2 different pH levels (pH 8.2 and 8.7). The pH of the experimental medium was modified with NaHCO_3 (1.8 mM NaHCO_3 for pH 8.2 and 5.6 mM NaHCO_3 for pH 8.7). Sodium concentrations were set equal between the 2 pH treatments with NaSO_4 . The pH during all experiments was regularly monitored and adjusted with diluted NaOH or HCl to target levels when pH deviated more than 0.1 pH units from target pH, to maintain a clear distinction between the pH 8.2 and 8.7 treatments. All test media were aerated for 2 d before spiking to allow equilibration. The Ni concentration series were prepared by adding NiCl_2 . The spiked test media were left to equilibrate for 1 d before test initiation. All chemicals were purchased from VWR International (reagent grade).

Ecotoxicity testing with *P. subcapitata*

The *P. subcapitata* 72-h growth inhibition test was conducted following the Organisation for Economic Co-operation and Development (OECD) guideline 201 [18]. A modified algal OECD medium was used (see *General design*). Furthermore, the Na_2 -ethylenediaminetetra-acetic acid (EDTA) in the original medium composition was replaced with dissolved organic carbon (DOC) originating from Aldrich Humic Acid (Sigma-Aldrich) with an equal C content (based on molecular mass), to keep Fe in solution in the OECD medium [4]. The medium composition is reported in Supplemental Data, Table S1.1. The algae originated from the Culture Collection

of Algae and Protozoa (CCAP 278/4). Nickel toxicity was investigated at 8 Ni concentrations (32–1800 μg nominal Ni/L) by comparison with a control (0 μg Ni/L; i.e., no Ni added) for each pH level.

A culture of the algae was set up 4 d before testing in aerated activated carbon-filtered Ghent city tap water to which the modified Provasoli's ES enrichment [19] at one-half strength was added, along with 1.4 mg/L $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, 15 mg/L $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$, 150 mg/L NaNO_3 , and 2.35 mg/L $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$. The flasks containing the algae were placed on a shaking device under continuous light (120 μmol photons $\text{m}^{-2} \text{s}^{-1}$) at 25 °C. The experiment was conducted in 100-mL Erlenmeyer flasks, which were filled with 50 mL of the modified OECD medium. Each concentration had 3 replicates and a blank correction (no algae). All replicates, except the blank corrections, were inoculated with 10^4 algae cells/mL. The flasks were incubated at 25 °C under continuous light (120 μmol photons $\text{m}^{-2} \text{s}^{-1}$). The pH of all replicates was measured at test initiation and also twice per day during the experiment. Algal cell density (cells/mL) was measured after 12 h, 48 h, and 72 h with a Beckman Z1TM series Coulter Counter cell and particle counter. Algal growth rate was calculated as the slope of $\ln(N_{it})$ versus t_i , where N_{it} is the cell concentration at time t_i and t is the time since the test initiation [18]. Test validity was evaluated as described by the OECD [18].

Ecotoxicity testing with *D. magna*

The *D. magna* 21-d reproduction test followed the OECD guideline 211 [20]. The test was conducted in modified M4-medium [21]. However, Na_2 -EDTA was omitted from the medium, and the medium contained 1.5 mM CaCl_2 , 0.25 mM MgSO_4 , and 1.8 mM NaHCO_3 (pH 8.2) or 5.6 mM NaHCO_3 (pH 8.7). The entire medium composition is summarized in Supplemental Data, Table S1.2.

Test animals originated from an in-house culture (clone K6), which has been maintained for more than 20 yr at 20 °C in activated carbon-filtered Ghent city tap water to which 1 μg /L Se, 75 μg /L thiamine, 1 μg /L cyanocobalamine, and 0.75 μg /L biotine are added. Daphnids were acclimated for 1 generation to the pH levels and the experimental medium to be tested. The toxicity test was initiated with juveniles from the third brood. Daphnids were 0 h to 24 h old at test initiation. The test was conducted in aquaria with 2 L content, and each aquarium contained 5 daphnids, each in a separate test unit. This test unit consisted of a polyvinylchloride (PVC) tube with gauze (200- μm mesh width) through which medium and algae, but not daphnids, could freely move. The experimental setup consisted of 2 aquaria per concentration and thus 10 replicates per Ni concentration and pH level in total. Nickel toxicity was investigated at 6 Ni concentrations (32–560 μg nominal Ni/L) and a control (0 μg Ni/L; i.e., no Ni added). The aquaria were placed in a growth chamber at 20 °C under a 16:8-h light: dark cycle. The test media were aerated and renewed twice per week during the experimental period (days 4, 7, 10, 13, and 17). The pH was measured daily and modified when needed. Daphnids were fed with a mixture of *P. subcapitata* and *Chlamydomonas reinhardtii* (3:1 proportion on a cell number basis) at renewal. The concentration of added algae was 5 mg dry wt/L during week 1, 10 mg/L during week 2, and finally 15 mg/L during week 3. Daphnid reproduction and mortality was checked daily. Total reproduction per daphnid was measured by counting the juveniles in each daphnid test unit. Test validity was evaluated as described by the OECD [20].

Ecotoxicity testing with *L. minor*

The *L. minor* 7-d growth inhibition test was conducted following OECD guideline 221 [22], using the Swedish Standard *Lemna* growth medium modified to contain 1.5 mM CaCl_2 , 0.25 mM MgSO_4 , and 1.8 mM NaHCO_3 (pH 8.2) or 5.6 mM NaHCO_3 (pH 8.7). No 3-(N-morpholino)propane-sulfonic acid (MOPS) buffer was used, and EDTA was replaced with Aldrich humic acid with identical carbon concentration (Supplemental Data, Table S1.3), to keep Fe in solution in the OECD medium [4].

The *L. minor* originated from an in-house culture. Plants were acclimated for 2 wk to the growth medium at the respective pH levels at 25 °C before test initiation. Eight test concentrations ranging from 10 μg total Ni/L to 560 μg total Ni/L were used, as well as a control treatment (no Ni added). There were 3 replicates per test concentration and pH level. Glass jars with 150 mL of the test medium were used. Each jar received 3 *L. minor* plants with 4 fronds each (in total, 12 fronds per jar and 36 fronds per treatment) at test initiation. The experiment was conducted in a 25 °C growth chamber under continuous light (470 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$). The medium was completely renewed on day 3 and day 5. The pH was measured daily and adjusted when needed. The number of fronds in each jar was counted on days 3, 5, and 7. The growth rate of *L. minor* was determined as the slope of $\ln(N_{ti})$ versus t_i , where N_{ti} is the number of fronds at time t_i and t_i is the time since the test initiation. Test validity was evaluated as described by the OECD [22].

Ecotoxicity testing with *B. calyciflorus*

The *B. calyciflorus* 48-h reproduction test was conducted following protocol 8420 of the American Public Health Association [23]. Rotokit F-test kits were purchased from MicroBioTests. A modified US Environmental Protection Agency medium was used for ecotoxicity testing (Supplemental Data, Table S1.4).

Cysts of the rotifers were hatched in 10 mL of the growth medium before test initiation at 25 °C on a light box. After hatching, rotifers were fed with a specific rotifer food (Roti-Rich food), which was supplied together with the rotifer test kit, for 2 h before transfer to the test solutions. Seven test concentrations ranging from 100 μg nominal Ni/L to 10 000 μg nominal Ni/L were used, as well as a control treatment (no Ni added). Algae (*P. subcapitata*) were added to the experimental medium in a concentration of 2×10^6 cells/mL. The rotifer test was conducted in two 24-well plates. To each well, 1 mL of the experimental medium was added. Each test concentration was replicated 10 times. One rotifer was placed in each well, after which the plates were sealed with Parafilm and placed in a dark 25 °C growth chamber. The number of rotifers was counted 48 h after test initiation, using a dissecting microscope. The population growth rate was calculated following the method described by De Schampelaere et al. [24]. Test validity was evaluated as described by Cleceri et al. [23].

Ecotoxicity testing with *L. stagnalis*

The *Lymnaea stagnalis* 28-d growth test was conducted following an in-house protocol. The composition of the experimental medium is given in Supplemental Data, Table S1.5.

Snails were hatched from egg strains originating from the breeding culture of the Vrije Universiteit Amsterdam (Amsterdam, The Netherlands). After hatching, snails were acclimated for 2 d to the experimental medium adjusted to the respective pH levels at 20 °C under a 12:12-h light:dark cycle.

Snails were 2 d to 3 d old at test initiation. Nickel toxicity was investigated at 7 concentrations ranging from 1 μg nominal Ni/L to 100 μg nominal Ni/L and a control treatment (no Ni added). The experiments were conducted in 2-L aquaria, and each aquarium contained 4 snails, each in a separate test unit. This test unit consisted of a PVC tube with gauze (200- μm mesh width) through which the medium, but not the snails, could move freely. The experimental setup consisted of 2 aquaria per concentration and thus 8 replicates per concentration and pH level in total. Snails were fed with 25 mg/L organic lettuce. The pH was measured daily and modified when needed. The medium and food were completely renewed 3 times per week. Shell length of the snails was measured on day 0 (before test initiation), day 14, and day 28 with a stereo microscope (SX40, Olympus), and the software cellSens[®] (Olympus). Shell length was defined as the longest dimension from the bottom of the shell to the top (Supplemental Data, Figure S1.1). The snail growth rate was calculated as the slope of $\ln(L_{ti})$ versus t_i , where L_{ti} is the shell length at time t_i and t_i is the time since the test initiation.

Chemical analysis

Samples of experimental media were taken at regular intervals (depending on the test species, but at least at the start and at the end of the test) during all experiments for analysis of total (only new medium) and dissolved (new and old medium; 0.45- μm filter, Acrodisc, PALL Life Sciences) Ni, organic carbon, and inorganic carbon. Samples for Ni analysis were acidified to 1% HNO_3 (Normat quality; VWR International) before measurement of Ni concentration by graphite furnace atomic absorbance spectroscopy (GFS, Thermo Fisher Scientific; Reference material TM-24.3, lot 0510 [Environment Canada]; limit of quantification 1 μg Ni/L, method detection limit 0.5 μg Ni/L) or flame atomic absorption spectrometry (SpectrAA800, Varian; Reference material TMDA-70, lot 0310 [Environment Canada]; limit of quantification 60 μg Ni/L, method detection limit 20 μg Ni/L) for Ni concentrations less than and greater than 100 μg /L, respectively. Dissolved organic carbon and dissolved inorganic carbon were analyzed with an Total Organic Carbon analyzer (TOC-5000, Shimadzu). The pH was measured with a pH glass electrode (P407, Consort), except for the pH at the end of the *B. calyciflorus* experimental period, for which pH was measured with a micro pH glass electrode (Knick).

Data analysis

Ni toxicity data were analyzed with STATISTICA 7 (Stat Soft). Effect concentrations (10%, 20%, and median effective concentrations [EC10, EC20, and EC50, respectively]) for each species and pH level were determined by fitting a 2-parameter log-logistic concentration response curve to the data based on the average measured dissolved Ni concentrations (arithmetic mean of new and old medium) [25]. Response curves were fitted to growth response data relative to control (%), except for the *D. magna* dataset, where response curves were fitted to the number of juveniles per female relative to the control (%). Data were checked for significant hormesis effects based on the equation reported by Van Ewijk and Hoekstra [26]. The Wheeler ratio-test was used to evaluate whether effect concentrations ($\text{EC10}_{\text{pH8.2}}$ vs $\text{EC10}_{\text{pH8.7}}$; $\text{EC20}_{\text{pH8.2}}$ vs $\text{EC20}_{\text{pH8.7}}$; $\text{EC50}_{\text{pH8.2}}$ vs $\text{EC50}_{\text{pH8.7}}$) at pH 8.2 and pH 8.7 were significantly different [27].

Bioavailability modeling

To investigate whether the existing chronic Ni bioavailability models can be reliably extrapolated to pH above 8.2,

predictions of Ni toxicity at pH 8.7 were made based on the observed toxicity at pH 8.2. First, speciation of Ni and other ions was calculated with the software package WHAM (Windermere Humic Aqueous Model) VI [28]. The default stability constants for inorganic carbonate complexes of Ni, Cu, and Zn in WHAM VI were adapted to those proposed by the National Institute of Standards and Technology (see Supplemental Data, Table S2). Background DOC concentrations and low additions of DOC (32 µg/L for *P. subcapitata* and 45 µg/L for *L. minor*) did not affect Ni speciation (Supplemental Data, Figure S3.1). Hence, fulvic acid and humic acid concentrations were set to zero for speciation modeling in WHAM VI. The medium composition used for speciation modeling is given in Supplemental Data, Table S3.1.

Predictions of Ni toxicity were made using the existing chronic Ni bioavailability models for *C. dubia* [5], *D. magna* [3], and *P. subcapitata* [4]. The *C. dubia* and *D. magna* models have an identical model structure, with equal biotic ligand (BL)-binding stability constants for Mg^{2+} and Ca^{2+} ($K_{MgBL, invertebrate}$ and $K_{CaBL, invertebrate}$) but differ in the magnitude of the pH effect ($S_{pH, C. dubia}$ or $S_{pH, D. magna}$; Equation 1 and Equation 2, respectively). The *P. subcapitata* bioavailability model consists of a log-linear pH effect that is superimposed on the single-site Mg^{2+} competition effect ("Alternative model" in Deleebeeck et al. [4]; Equation 3).

$$ECx_{Ni^{2+}, i}^{pred} = 10^{-(Q_{x, C. dubia} + S_{pH, C. dubia} \times pH_i)} \times \left\{ 1 + K_{MgBL, invertebrate} \times \{Mg^{2+}\}_i + K_{CaBL, invertebrate} \times \{Ca^{2+}\}_i \right\} \quad (1)$$

$$ECx_{Ni^{2+}, i}^{pred} = 10^{-(Q_{x, D. magna} + S_{pH, D. magna} \times pH_i)} \times \left\{ 1 + K_{MgBL, invertebrate} \times \{Mg^{2+}\}_i + K_{CaBL, invertebrate} \times \{Ca^{2+}\}_i \right\} \quad (2)$$

$$ECx_{Ni^{2+}, i}^{pred} = 10^{-(Q_{x, P. subcapitata} + S_{pH, P. subcapitata} \times pH_i)} \times \left\{ 1 + K_{MgBL, P. subcapitata} \times \{Mg^{2+}\}_i \right\} \quad (3)$$

In these models $ECx_{Ni^{2+}, i}^{pred}$ is the predicted Ni^{2+} activity for the $x\%$ effect concentration in test solution i ; $K_{MgBL, invertebrate}$ and $K_{CaBL, invertebrate}$ are the stability constants for binding of Mg^{2+} and Ca^{2+} to the biotic ligand of the invertebrates (*D. magna* and *C. dubia*), respectively, and $K_{MgBL, P. subcapitata}$ is the stability constant for binding of Mg^{2+} to the *P. subcapitata* biotic ligand; $\{Mg^{2+}\}_i$ and $\{Ca^{2+}\}_i$ are the chemical activities of Mg^{2+} and Ca^{2+} in test solution i (mol/L), respectively; and $Q_{x, D. magna}$, $Q_{x, C. dubia}$ and $Q_{x, P. subcapitata}$ are the specific intrinsic sensitivities of the considered species using the *D. magna*, *C. dubia*, and *P. subcapitata* bioavailability models for Ni, respectively. The intrinsic sensitivity is the intercept of the linear relationship between $pEC50_{Ni^{2+}}^*$ (i.e., the negative logarithm of $EC50_{Ni^{2+}}$ after correction for Ca^{2+} or Mg^{2+} competition) and pH [3,4]. Finally, $S_{pH, D. magna}$, $S_{pH, C. dubia}$, and $S_{pH, P. subcapitata}$ are the pH slopes of these linear relationships for the respective model organisms *D. magna*, *C. dubia*, and *P. subcapitata*, and pH_i is the pH of test solution i . All model parameters are listed in Table 1.

The following method was used for bioavailability modeling: In a first step, the bioavailability models were used to calculate an intrinsic sensitivity (Q_x) based on the Ni toxicity

Table 1. Model parameters of chronic Ni bioavailability models for *Daphnia magna* (Equation 3 [3]), *Ceriodaphnia dubia* (Equation 2 [5]) and *Pseudokirchneriella subcapitata* (Equation 4 [4]) and intrinsic sensitivities for the 5 species tested in this study calculated based on the observed Ni^{2+} toxicity at pH 8.2

Parameter	Species group		
	Invertebrate models		Plant model
	<i>D. magna</i> model	<i>C. dubia</i> model	<i>P. subcapitata</i> model
Log K_{CaBL}	3.53	3.53	—
Log K_{MgBL}	3.57	3.57	3.3
S_{pH}^a	0.335 ± 0.055	0.859 ± 0.200	0.143 ± 0.052
Intrinsic sensitivities			
$Q_{50, D. magna}$	3.34	—	—
$Q_{20, D. magna}$	3.76	—	—
$Q_{50, Brachionus calyciflorus}$	2.66	−1.59	—
$Q_{20, B. calyciflorus}$	2.77	−1.49	—
$Q_{20, Lymnaea stagnalis}$	4.54	0.29	—
$Q_{50, P. subcapitata}$	—	—	4.64
$Q_{20, P. subcapitata}$	—	—	4.85
$Q_{50, Lemna minor}$	—	—	4.63
$Q_{20, L. minor}$	—	—	5.22

^aFor slopes of the effect of pH on chronic Ni toxicity (S_{pH}), the mean $\pm$ standard deviation is reported for *D. magna* and *C. dubia*. Standard deviation was estimated based on the data reported in Deleebeeck et al. [3] and De Schampelaere et al. [5], respectively.

K_{CaBL} = stability constant for binding of Ca^{2+} to the biotic ligand; K_{MgBL} = stability constant for binding of Mg^{2+} to the biotic ligand; Q_x = specific intrinsic sensitivity of the considered species.

at pH 8.2 (i.e., using the effect concentrations at pH 8.2). A species-specific intrinsic sensitivity was calculated for each species individually, because Ni sensitivities differ between species. In a second step, this intrinsic sensitivity was then used to predict Ni toxicity ($EC50s$ or $EC20s$, expressed as Ni^{2+} activity) at pH 8.7. In a final step, predicted Ni^{2+} activities were translated to dissolved Ni concentrations with WHAM VI. All predicted effect concentrations were compared with the observed effect concentrations at pH 8.7.

Because species-specific bioavailability models are available for *P. subcapitata* and *D. magna*, it is obvious to use these species-specific models for the prediction of toxicity data for these 2 species. For the nonmodel invertebrates *L. stagnalis* and *B. calyciflorus*, both the *D. magna* and *C. dubia* models were used to predict chronic Ni toxicity. Finally, the *P. subcapitata* model was used for predictions of Ni toxicity to the plant *L. minor*. The pH slopes of observed and predicted data were compared by plotting the observed and predicted Ni toxicity against pH.

To compare the observed pH slopes of the different tested species with the pH slopes of the chronic Ni bioavailability models, toxicity data ($EC20_{Ni^{2+}}$ and $EC50_{Ni^{2+}}$) were hardness corrected (i.e., corrected for the presence of Ca or Mg) using Equation 4 for invertebrates and using Equation 5 for algae and higher plants. The pH slopes (S_{pH}) were determined for every species based on the regression equation of the linear relationship between the logarithm of hardness-corrected ECx and the pH at 8.2 and 8.7.

$$ECx_{Ni^{2+}, hardness\ corr, invertebrates, i} = \frac{ECx_{Ni^{2+}, Observed, i}}{\left\{ 1 + K_{MgBL, invertebrate} \times \{Mg^{2+}\}_i + K_{CaBL, invertebrate} \times \{Ca^{2+}\}_i \right\}} \quad (4)$$

$$ECx_{Ni^{2+}, hardness\ corr, plants, i} = \frac{ECx_{Ni^{2+}, Observed, i}}{\{1 + K_{MgBL, P. subcapitata} \times \{Mg^{2+}\}_i\}} \quad (5)$$

In these models, $ECx_{Ni^{2+}, hardness\ corr, invertebrates, i}$ and $ECx_{Ni^{2+}, hardness\ corr, plants, i}$ are the hardness corrected $x\%$ effect concentration in test solution i for invertebrates and plants, respectively, and $ECx_{Ni^{2+}, observed, i}$ is the observed $x\%$ effect concentration in test solution i , expressed as Ni^{2+} activity. The standard deviation of S_{pH} was determined through bootstrapping [29]. For every species, 5000 random samples of the EC50s at both pH levels were drawn from a normal distribution defined by the estimated mean and standard error of the hardness corrected EC50s. These 5000 sampled EC50 pairs were then used to calculate 5000 values of S_{pH} for every species, from which the standard deviation of S_{pH} was calculated. Five thousand sample pairs were taken, because this number resulted in an adequate estimation of the variability of the pH slopes. The $S_{pH \leq 8.2}$ and $S_{pH > 8.2}$ were statistically compared for the bioavailability model species *P. subcapitata*, *D. magna*, and *C. dubia* using the method described by Sokal and Rohlf [30].

RESULTS

Effect of pH on chronic Ni toxicity

The validity criteria of the standard tests were met for most species (Supplemental Data, Table S4.1). Concentration–response data and fitted concentration–response curves for the 5 species based on dissolved Ni concentrations and Ni^{2+} activities are shown in the Supplemental Data (Figures S4.1–5). Dissolved Ni and free Ni^{2+} toxicity for the 5 tested species, expressed as EC10, EC20, and EC50, are summarized in

Table 2. The EC50 of the pH 8.7 treatment for *L. minor* and the EC50 of the pH 8.2 treatment for *D. magna* were extrapolated but were within a factor of 1.5 from the highest tested concentration. No reliable EC50 for the pH 8.2 treatment of *L. stagnalis* could be calculated, because only 25% effect at the highest tested concentration was observed. Dissolved Ni toxicity was significantly higher (i.e., lower ECx) at pH 8.7 than at pH 8.2 for *L. stagnalis*, *P. subcapitata*, and *B. calyciflorus*. For *D. magna*, dissolved Ni toxicity was higher at pH 8.7 than at pH 8.2; however, this difference was only significant at the EC50 level. For *L. minor*, differences in effect concentrations between pH 8.2 and 8.7 were not significant. The largest relative difference in dissolved Ni toxicity was observed for snails: Ni toxicity at the level of EC10 and EC20 was more than 3-fold higher at pH 8.7 than at pH 8.2. The difference in effect concentrations between pH 8.2 and 8.7 was more distinct when expressed on the basis of Ni^{2+} activity. The effect concentrations expressed as Ni^{2+} activity were for all species significantly lower at pH 8.7 than at pH 8.2 (i.e., more toxic), except for the EC50 of *L. minor*. Effect concentrations expressed as Ni^{2+} activities were at least 1.9-fold higher (i.e., less toxic) at pH 8.2 than at pH 8.7. Again, the largest difference in effect concentrations was observed for *L. stagnalis*: EC10 $_{Ni^{2+}}$ and EC20 $_{Ni^{2+}}$ were more than 5-fold higher at pH 8.2 than at pH 8.7.

Prediction of Ni toxicity at pH 8.7

Predictions of Ni toxicity at pH 8.7 for the different species were made based on observed Ni toxicity at pH 8.2. More specifically, the effect concentrations of the pH 8.2 treatments listed in Table 2 were used to calculate species-specific intrinsic sensitivities (Qx in Table 1). The EC20 and EC50 were used to make predictions for *P. subcapitata*, *L. minor*, *B. calyciflorus*, and *D. magna*, whereas only the EC20 was used for *L. stagnalis*.

Table 2. Measured pH and Ni toxicity (EC10, EC20, and EC50), expressed as dissolved Ni concentrations and calculated Ni^{2+} activity, for all species at both pH levels^a

Species	pH ± SD	EC10 (dissolved Ni µg/L)	EC20 (dissolved Ni µg/L)	EC50 (dissolved Ni µg/L)	EC10 (Ni^{2+} activity nmol/L)	EC20 (Ni^{2+} activity nmol/L)	EC50 (Ni^{2+} activity nmol/L)
<i>Pseudokirchneriella subcapitata</i>	8.17 ± 0.03	132 (118–148)	175 (162–190)	285 (272–299)	926 (827–1037)	1230 (1140–1329)	2000 (1905–2099)
	8.65 ± 0.07	75* (64–89)	106* (93–119)	188* (174–203)	254* (216–301)	357* (315–402)	634* (587–686)
<i>Lemna minor</i>	8.21 ± 0.15	28 (19–57)	73 (52–102)	286 (208–394)	229 (132–398)	512 (364–718)	2010 (1461–2770)
	8.68 ± 0.07	28 (15–52)	75 (52–108)	404 ^a (240–681)	94* (50–176)	252* (175–360)	1356 ^b (805–2284)
<i>Daphnia magna</i>	8.16 ± 0.07	109 (36–331)	191 (107–339)	492 ^a (190–1271)	800 (263–2427)	1398 (787–2483)	3606 ^a (1394–9341)
	8.67 ± 0.06	94 (39–228)	107 (52–221)	134* (84–212)	412 (170–997)	469* (228–963)	584* (369–925)
<i>Brachionus calyciflorus</i>	8.12 ± 0.05	1576 (1244–2194)	1913 (1538–2379)	2459 (2128–2840)	11379 (8967–15 826)	13793 (11 083–17 139)	17732 (15 360–20 487)
	8.55 ± 0.18	1098* (990–1219)	1194* (1078–1324)	1576* (1369–1815)	5078* (4589–5650)	5514* (4997–6140)	7311* (6330–8401)
<i>Lymnaea stagnalis</i>	8.11 ± 0.04	19 (8.5–44.7)	32 (21–50)	>39 ^b	146 (64–334)	241 (153–377)	>291 ^c
	8.64 ± 0.07	4.6* (2.2–9.6)	10.2* (6.5–16.2)	40.1 (25.9–62.2)	21* (10–44)	47* (30–75)	186 (120–287)

^a95% confidence intervals are reported between parentheses.

^bEC50 slightly extrapolated, but within a factor 1.5 of highest tested concentration.

^cNo reliable EC50 could be determined, because less than 50% reduction was observed at the highest tested concentration. The highest tested concentration/activity is reported instead.

*Indicates that the ECx for a species at pH 8.7 is significantly lower than at pH 8.2 according to the Wheeler ratio-test [27].

SD = standard deviation; EC10 = 10% effective concentration; EC20 = 20% effective concentration; EC50 = median effective concentration.

Generally, Ni toxicity at pH 8.7 was underestimated (i.e., ECx overestimated) using the existing Ni bioavailability models for all effect concentrations and for all species (almost all points above the 1:1 line; Figure 1 upper panel). For *P. subcapitata* and *D. magna*, for which species-specific bioavailability models are available, EC20_{Nidiss} were overestimated, with 3.0-fold and 2.2-fold error, respectively, and EC50_{Nidiss} were overestimated, with 2.7-fold and 4.6-fold error, respectively. The *P. subcapitata* model predicted Ni toxicity to *L. minor* quite accurately: EC20_{Nidiss} and EC50_{Nidiss} were predicted within 2-fold error of the observed ones. For *L. stagnalis*, EC20_{Nidiss} was overestimated by the *D. magna* model, with 3.7-fold error. For *B. calyciflorus*, EC20_{Nidiss} and EC50_{Nidiss} were overestimated by the *D. magna* model, with 2-fold and 2.2-fold error, respectively. However, the *C. dubia* model predicted Ni toxicity to those invertebrates more accurately than the *D. magna* model.

Observed and predicted logEC20 values, expressed as Ni²⁺ activity, are plotted as a function of pH for the different species in Figure 2. Similar figures for the EC50 of *P. subcapitata* and *B. calyciflorus* are provided as Supplemental Data, Figure S5.1. The observed pH slope between 8.2 and 8.7 is steeper than the pH slope predicted by the chronic Ni bioavailability models. The *P. subcapitata* model clearly underestimated the pH slope for both *P. subcapitata* and *L. minor*. The pH

slopes of the observed data were 1.12 and 0.66 for the algae and the macrophyte, respectively. The predictions of the *D. magna* model failed to reflect the observed change in Ni toxicity with increasing pH for *D. magna*. Indeed, the observed pH slope for *D. magna* above pH 8.2 was 0.93, whereas that of the *D. magna* model is 0.34. For both *B. calyciflorus* and *L. stagnalis*, the *D. magna* model underestimated the pH slope of Ni toxicity, whereas the *C. dubia* model predicted the change in Ni toxicity when increasing the pH relatively well. An overview of hardness-corrected pH slopes (S_{pH}) is shown in Table 3 and Figure S5.2. The S_{pH} of the existing chronic Ni bioavailability models (Table 1) were on average (i.e., compared in group: S_{pH} below pH 8.2 vs S_{pH} above pH 8.2 of *P. subcapitata*, *D. magna*, and *C. dubia*) significantly smaller than the S_{pH} between pH 8.2 and 8.7 of the same model species (one-sided *t* test: $p = 0.03$, $n = 3$). Additionally, for each of the bioavailability model species *P. subcapitata*, *D. magna*, and *C. dubia*, the S_{pH} was significantly steeper between pH 8.2 and 8.7 than below pH 8.2 (*F*-test [26], $p \leq 0.001$ in all cases; Figure 3).

DISCUSSION

Effect of high pH on chronic Ni toxicity

The present study investigated the effect of increasing the pH from 8.2 to 8.7 on chronic Ni toxicity. Previously, pH has been shown to be an important water chemistry variable in determining chronic Ni toxicity below pH 8.2; in general, chronic Ni toxicity below pH 8.2 increases with increasing pH for fish, daphnids, and algae [2–5,31]. The present study extends the observed trend to higher pH levels: chronic dissolved Ni toxicity at pH 8.7 was almost always higher than at pH 8.2. The difference in Ni toxicity between both pH levels is even larger at the Ni²⁺ activity level. This is a consequence of speciation: When pH increases, inorganic Ni complexes, such as Ni-hydroxides and Ni-carbonates, will gradually become more dominant, resulting in a lower concentration of free Ni²⁺ ions than at lower pH [32]. The mechanism of how pH influences Ni toxicity is currently not well understood. Although the relation between cations that compete with metals for binding at the biotic ligand site and free metal ion toxicity is in general linear (e.g., Mg²⁺ for Ni²⁺ toxicity [2–4] and H⁺ for Pb²⁺ toxicity to *C. dubia* [25]), the relationship between H⁺ and Ni²⁺ toxicity is rather nonlinear. This suggests that the effect of H⁺ on chronic Ni²⁺ toxicity is not only attributable to the competition between H⁺ and Ni²⁺ for binding at a single monodentate biotic ligand site [2–4]. This nonlinearity may result from inorganic Ni-carbonate complexes, NiCO₃ and NiHCO₃⁺, which might be bioavailable and therefore also contribute to chronic Ni toxicity [33]. According to Deleebeeck et al. [2,3], however, this is unlikely because the toxicity of NiCO₃ and NiHCO₃⁺ would in that case be substantially higher than the toxicity of the free Ni²⁺ ion. Those authors instead suggested that Ni²⁺ may dissociate from Ni-carbonate complexes in the physicochemically different microenvironments surrounding the biotic ligand, leading to an increase in bioavailable Ni²⁺. Alternatively, because Ni²⁺ is most likely a Mg²⁺ antagonist [34], Ni-carbonate complexes may potentially enhance uptake of Ni²⁺ through a Mg²⁺/HCO₃⁻ symporter, which has previously been characterized in tumor cells [35]. Additionally, a difference in pH can possibly alter the cell membrane integrity [36]. Finally, a nonlinear relation between H⁺ and Ni²⁺ also may indicate the existence of more than 1 type of Ni²⁺ binding site with different protonation constants, as was proposed for Cu toxicity to the amphipod *Hyalella azteca* [37].

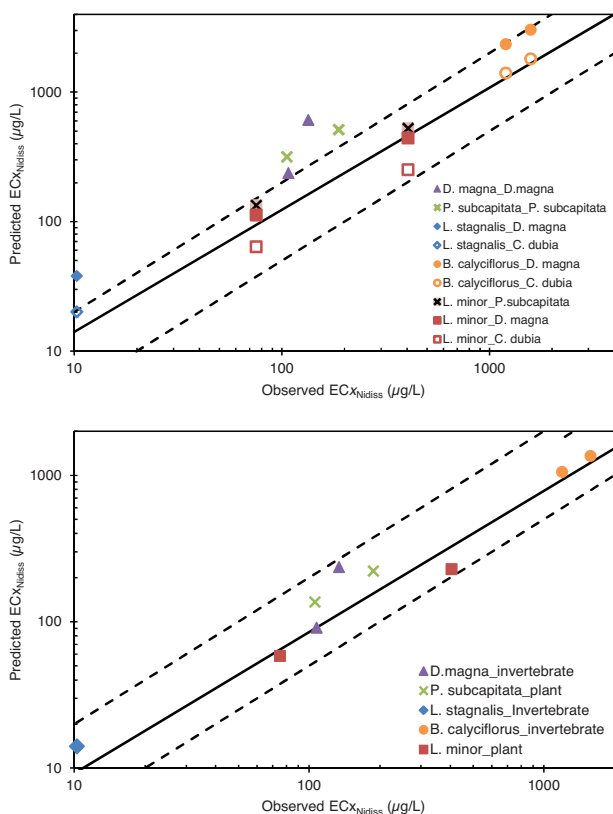


Figure 1. Observed versus predicted median and 20% effective concentrations (EC50 and EC20), expressed as dissolved Ni/L, at pH 8.7 for the different tested species. The dashed lines represent a difference of a factor of 2 between the observed and predicted data. The full line represents a perfect fit between observed and predicted data. Predictions in the upper panel were made based on chronic Ni toxicity at pH 8.2 using Equation 2 for the *C. dubia* model, Equation 3 for the *D. magna* model, and Equation 4 for the *P. subcapitata* model. Predictions in the lower panel were made with the bioavailability models for pH above 8.2 developed in the present study. Speciation was calculated with WHAM VI. Legend is denoted as follows: Species bioavailability model.

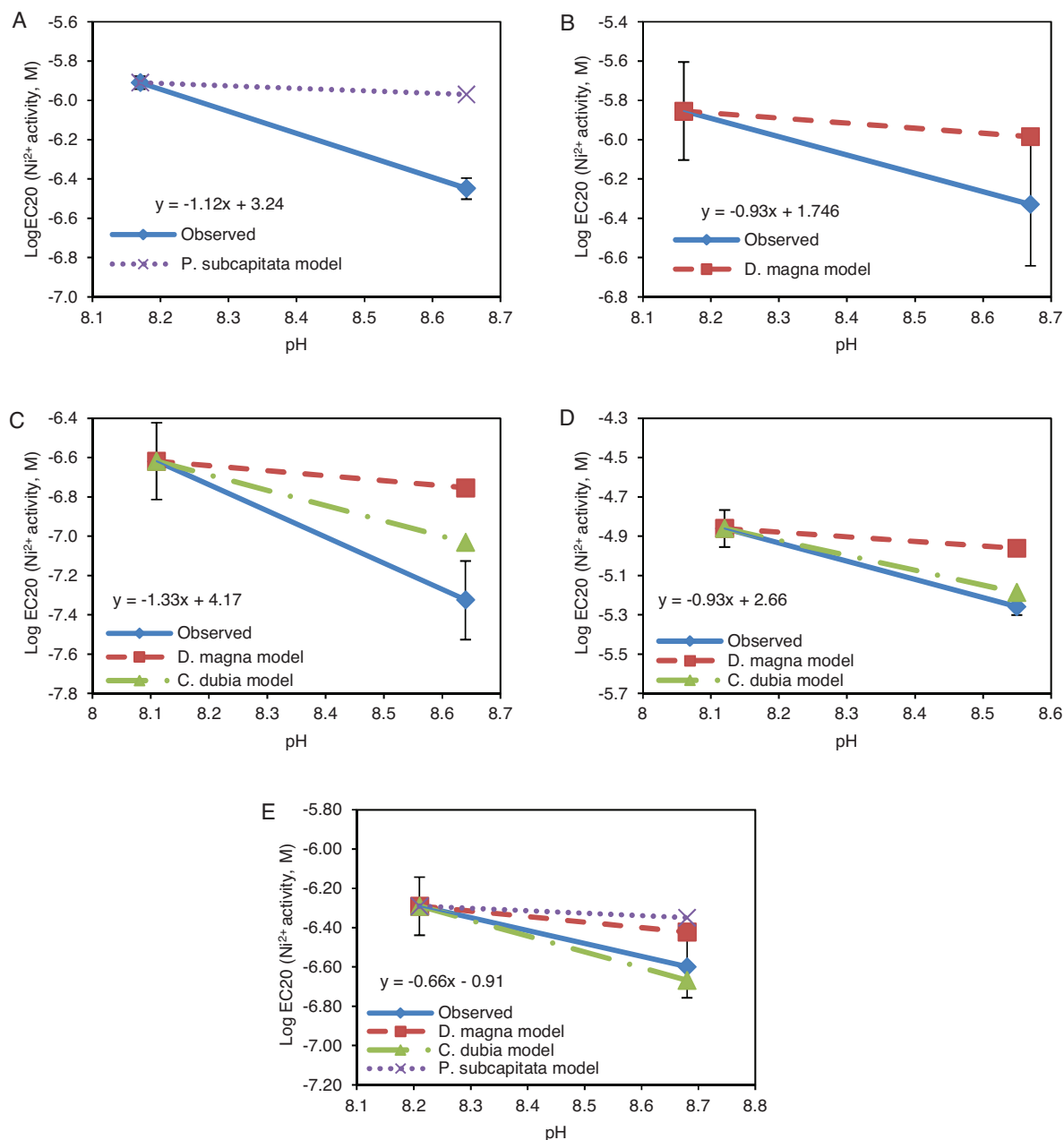


Figure 2. Observed and predicted log₁₀ 20% effective concentration ($EC_{20, Ni^{2+}}$) as a function of pH for (A) *Pseudokirchneriella subcapitata*, (B) *Daphnia magna*, (C) *Lymnaea stagnalis*, (D) *Brachionus calyciflorus*, and (E) *Lemna minor*. Plotted error bars indicate 95% confidence intervals on EC_{20} . The equation of the linear relationship between pH and observed Ni^{2+} toxicity is reported. Lines are denoted as follows: observed toxicity (full line), *D. magna* model for pH up to 8.2 predicted toxicity (dashed line), *Ceriodaphnia dubia* model for pH up to 8.2 predicted toxicity (dashed-dotted line), and *P. subcapitata* model for pH up to 8.2 predicted toxicity (dotted line).

Because of the nonlinear effect of H^+ on chronic Ni^{2+} toxicity, the effect of pH in the chronic Ni bioavailability models was not modeled using a classic biotic ligand model-type single-site competition effect of H^+ , which is typically denoted by the stability constant for binding of H^+ to the biotic ligand: K_{HBL} [7,38]. Instead, empirical regressions in which the log-linear effect of pH on Ni^{2+} toxicity was superimposed on the single-site competition effects of Mg^{2+} or Ca^{2+} were used, denoted by a pH slope: S_{pH} (Equations 1–3 [2–5]).

Validation of existing chronic Ni bioavailability models at pH > 8.2

We investigated whether the existing chronic Ni bioavailability models, which are validated for a pH ≤ 8.2 , also could be used to predict Ni toxicity at pH 8.7 based on the observed Ni

toxicity at pH 8.2 for 5 species, by calibrating the intrinsic sensitivity on the effect concentrations at pH 8.2. Chronic Ni toxicity to *L. minor* and *B. calyciflorus* at pH 8.7 was predicted with reasonable accuracy (11 of 15 within 2-fold error). Our results overall suggest that the existing bioavailability models cannot be safely extrapolated above pH 8.2, because Ni toxicity predicted at pH 8.7 (as extrapolated from observed toxicity at pH 8.2) was underestimated for several species (many points above the 1:1 line in Figure 1). For instance, for the model species *D. magna* and *P. subcapitata*, for which we expect that the species-specific Ni bioavailability models predict Ni toxicity within 2-fold error [3,4], prediction errors at the level of the EC_{50} were 4.6-fold and 2.7-fold, respectively. It should, of course, be kept in mind that the EC_{50} of the pH 8.2 treatment of

Table 3. Slopes of the relation between log EC20_{Ni2+,hardnesscorrected} and log EC50_{Ni2+,hardnesscorrected} and pH (S_{pH}) for the different species^a

	$S_{pH,EC20}$	$S_{pH,EC50}$
Invertebrates		
<i>Lymanaea stagnalis</i>	1.410 ± 0.541	^c
<i>Daphnia magna</i>	1.010 ± 0.778	^d
<i>Brachionus calyciflorus</i>	1.027 ± 0.277	0.996 ± 0.197
<i>Ceriodaphnia dubia</i> ^b	—	1.878 ± 0.076
Mean invertebrates	1.149	1.438
Plants		
<i>Lemna minor</i>	0.673 ± 0.469	^d
<i>Pseudokirchneriella subcapitata</i>	1.139 ± 0.122	1.058 ± 0.076
Mean plants	0.906	1.058

^aMean value ± standard deviation is reported.^bData from Wirtz et al. [42], reported in De Schampelaere et al. [5]^cNo reliable EC50 could be determined for *L. stagnalis* at pH 8.2.^dEC50 of *D. magna* and *L. minor* were extrapolated and as such not incorporated in the pH slope analysis.

D. magna was somewhat extrapolated outside the tested concentration range and that there is therefore uncertainty associated with the predictions based on this EC50. At the highest tested concentration of the pH 8.2 treatment (311 µg/L), however, 38% effect on daphnid reproduction was observed, which for simplicity can be assumed as the EC38 (38% effect concentration). The predicted EC38 at pH 8.7, based on this EC38 value, is 387 µg/L. Yet, this predicted concentration is still 2.9-fold higher than the observed EC50 at pH 8.7, confirming that, independent of the uncertainty regarding the EC50 at pH 8.2, the *D. magna* bioavailability model in its current form underestimates chronic Ni toxicity to *D. magna* at pH 8.7. Figure 2 clearly shows the reason for the failed extrapolation of the chronic Ni bioavailability models. The observed pH slopes are always steeper than the predicted pH slopes. Indeed, the pH slopes of the existing chronic Ni bioavailability models for *P. subcapitata*, *D. magna*, and *C. dubia* (validated for a pH up to 8.2) range between 0.14 and 0.86 (Table 1), whereas the observed pH slopes between pH 8.2 and 8.7 in the present study ranged from 0.67 to 1.41 (Table 3). The pH slopes for *P. subcapitata* and *D. magna* above pH 8.2 were considerably steeper than the pH slopes of their respective species-specific bioavailability models; 1.139 versus 0.143 and 1.010 versus 0.335 (Figure 3). Similarly, also for *C. dubia*, a clear difference is seen in the size of the pH effect on Ni²⁺

toxicity below and above pH 8.2: the pH slope is 0.86 below pH 8.2 and 1.87 above pH 8.2 [5]. For *B. calyciflorus* and *L. stagnalis*, the *C. dubia* model approaches the observed effect of pH on Ni toxicity better than the *D. magna* model. Indeed, Ni toxicity to these invertebrates was most accurately predicted with the *C. dubia* model (Figure 1), although Schlegel et al. [10] found that Ni toxicity below pH 8.2 to *B. calyciflorus* was more accurately predicted with the *D. magna* model. The pH slope of the *C. dubia* model is more than twice ($S_{pH,pH \leq 8.2} = 0.8587$) the pH slope of the *D. magna* model ($S_{pH,pH \leq 8.2} = 0.335$). This explains why the *C. dubia* model predicts Ni toxicity to invertebrates above pH 8.2 better. However, the pH slopes above pH 8.2 for all invertebrate species were still steeper than the slope for pH ≤ 8.2 of the *C. dubia* model (Table 3). For *L. minor*, although toxicity was underestimated by the *P. subcapitata* model, predictions were still within a 2-fold error of observed toxicity. Nevertheless, the observed pH slope of *L. minor* was considerably higher than the pH slope of the *P. subcapitata* model. Because Schlegel et al. [10] suggested that the *D. magna* and *C. dubia* models predicted Ni toxicity to *L. minor* more accurately than the algae models, these invertebrate models also were used to predict Ni toxicity to this hydrophyte. These invertebrate models indeed gave more accurate predictions of the EC20 (Figure 1). However, the *D. magna* model still underestimated the observed pH slope, whereas the *C. dubia* model overestimated the pH slope (Figure 2E).

The results of the present study suggest that in general the pH slopes used in the existing chronic Ni bioavailability models are not steep enough to accurately predict toxicity to most species in water with a pH greater than 8.2. Furthermore, for the species for which a species-specific bioavailability model exists, the pH slopes for pH < 8.2 were significantly lower than those for pH ≥ 8.2. Hence, the slope of the pH function is dependent on the pH itself, and 2 distinct regions may be recognized: a slope lower than 1 below pH 8.2 and a slope close to or higher than 1 above pH 8.2. Different slopes of the pH function in different pH regions may suggest the presence of at least 2 biotic ligand sites, which both may be singly or doubly protonated [37]. The presence of 2 biotic ligand sites for Ni²⁺ has been suggested previously for *Hordeum vulgare* [39] and *Escherichia coli* [40], which suggests that this may be ubiquitous among organisms.

Development of chronic Ni bioavailability models for pH above 8.2

As discussed (see Validation of existing chronic Ni bioavailability models at pH > 8.2), the existing Ni bioavailability models cannot simply be extrapolated to predict Ni toxicity in waters above pH 8.2. Therefore, we modified the existing chronic Ni models by adapting the pH slope of the existing bioavailability models, depending on the pH of the target water for which toxicity should be predicted. Because both crustacean models have the same biotic ligand-binding constants for Ca and Mg (Table 1), a unified invertebrate bioavailability model was developed that contains an averaged invertebrate pH slope for pH above 8.2 (i.e., $S_{pH>8.2,invertebrates}$) and the Ca²⁺ and Mg²⁺ competition effects of the invertebrate models developed for pH below pH 8.2 (Equation 2). Likewise, a unified plant model was developed with an averaged plant pH slope for pH above 8.2 (i.e., $S_{pH>8.2,plant}$) and the Mg²⁺ competition effects of the *P. subcapitata* model (Equation 3). Table 3 lists $S_{pH>8.2}$ for the different species. For *B. calyciflorus* and *P. subcapitata*, the 2 species for which an EC20 and an EC50 at both pH levels could be accurately calculated (Table 2), the observed pH slopes were very similar independent of the

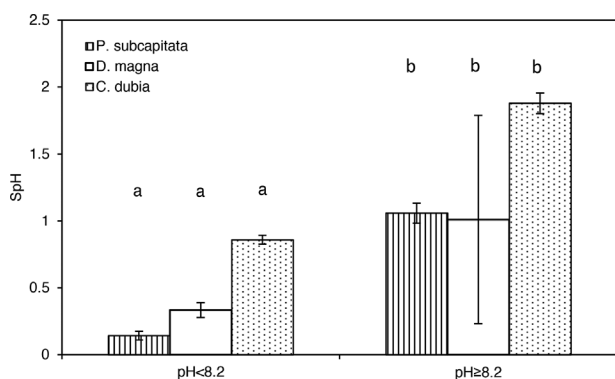


Figure 3. Slopes of the effect of pH on chronic Ni toxicity (S_{pH}) to *Pseudokirchneriella subcapitata* (striped bar), *Daphnia magna* (open bar), and *Ceriodaphnia dubia* (dotted bar) for pH < 8.2 (left, data from [3–5]) and pH ≥ 8.2 (right, *C. dubia* data from [5]). Plotted error bars represent standard deviations. Letters indicate whether S_{pH} below pH 8.2 and above pH 8.2 differ significantly for a species.

effect level (EC20 vs EC50) at which they were derived (Table 3). Therefore, the mean pH slope based on the EC20, for which more data are available, can be used to predict both EC20s and EC50s. Nickel toxicity for all invertebrate species was predicted within 2-fold error when the averaged invertebrate slope for pH greater than 8.2 ($S_{\text{pH, invertebrate}}$) of 1.149 was used, without the bias observed using the bioavailability models developed for pH below 8.2 (Figure 1 lower panel). Also, Ni toxicity to *L. minor* and *P. subcapitata* was more accurately predicted using the model with the averaged plant slope for pH greater than 8.2 ($S_{\text{pH, plant}} = 0.906$). The derivation of the pH slopes in the present study was of course based on only 2 data points for each species. As a consequence, our analysis implicitly assumes that the relationship between pH and Ni^{2+} toxicity is linear between pH 8.2 and 8.7.

Although Ni toxicity would be underestimated by the existing chronic Ni bioavailability models in the waters with pH above 8.2, which are common among European surface waters, the invertebrate and plant Ni bioavailability models for pH above 8.2 developed in the present study could result in a more accurate risk assessment of Ni toxicity in these waters.

Implications for risk assessment

Presently, the full bioavailability normalization approach is applied in Ni risk assessment procedures in the European Union [11]. This approach starts with a database that contains chronic ecotoxicity data for 31 different freshwater species (214 chronic data points in total), which is then normalized by using the existing chronic Ni bioavailability models to a set of 7 water types representing the typical characteristics of freshwater bodies in the European Union (i.e., “the ecoregions”) and to calculate a water type–specific hazardous concentration at the 5th percentile (HC5). Because the existing chronic Ni bioavailability models are only valid up to pH 8.2 and 15% of European surface waters have a pH above 8.2 [15], assessment of the risks coupled with Ni exposure are currently uncertain for a substantial fraction of surface waters in the European Union. Extrapolating the current bioavailability models to pH above 8.2 would lead to underestimations of risks, because the present study showed that the effect of pH on chronic Ni toxicity increases above pH 8.2. Therefore, we suggest modifying the bioavailability normalization approach currently used in the Ni risk assessment procedures when waters with pH above 8.2 are to be evaluated. For waters with pH above 8.2, we propose a 2-step procedure to normalize toxicity data (and calculate HC5) to a target water *t* with pH_t (> 8.2), DOC_t , and hardness_{*t*}: first, normalize all toxicity data first to pH 8.2, DOC_t , and hardness_{*t*}, using the bioavailability models that have been shown valid for pH up to 8.2 [2–5] as explained in Schlegel et al. [10]; and second, further extrapolate the toxicity data normalized to pH 8.2 in step 1 to pH_t , using a mean invertebrate $S_{\text{pH}} = 1.149$ for all invertebrates and a mean $S_{\text{pH}} = 0.906$ for all plants and algae (present study). The toxicity data from step 2 can subsequently be used to fit a species sensitivity distribution and calculate an HC5. Normalization of the toxicity data for a target water *t* with $\text{pH}_t < 8.2$, DOC_t , and hardness_{*t*}, is performed in a single-step procedure with the chronic Ni bioavailability models developed for pH < 8.2 . This approach is realized in an Excel environment: the “chronic Ni bioavailability and normalization tool” conducted in Supplemental Data, Section 6. For illustrative purposes, this “chronic Ni bioavailability and normalization tool” was used to calculate HC5 for one of the water bodies from the set of ecoregions. For this exercise, a water with pH close to the upper application boundary of the

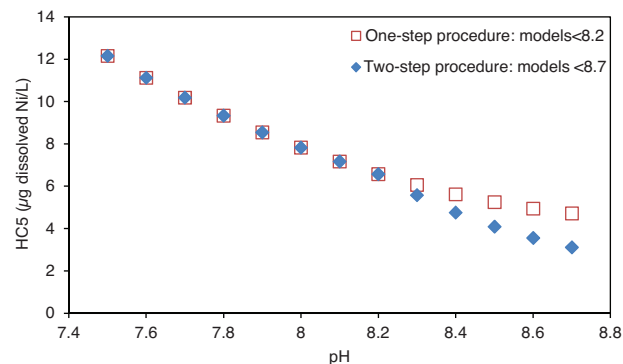


Figure 4. Calculated hazardous concentration at the 5th percentile (HC5) for the River Otter (United Kingdom; dissolved organic carbon = 3.2 mg/L; hardness = 165 mg CaCO_3/L) at different pH. Results for the one-step procedure used in the Ni risk assessment procedure [10,11] (using the bioavailability models validated for pH up to 8.2 [2–5], squares) and for the two-step procedure as proposed in the present study (diamonds).

existing Ni bioavailability models was chosen (Otter, UK; pH = 8.1). The pH of this river naturally fluctuates within a pH range of 7.1 to 8.7. For every 0.1 pH unit within this pH range, an HC5 was calculated with the 2-step procedure proposed in the present study, using the log-normal species sensitivity distribution [41]. The detailed water chemistry used for this bioavailability normalization exercise is provided in Supplemental Data, Table S7. For comparison, HC5 values were also calculated with the bioavailability normalization approach used in the Ni risk assessment procedure [10] (Supplemental Data, Section 6). The calculated HC5 values are the same for both approaches for pH ≤ 8.2 , because both approaches use the same models below or equal to pH 8.2 (Figure 4). Calculated HC5 starts to deviate at higher pH values. The deviation between both procedures increases when the pH increases. Although the calculated HC5 of the 2-step procedure is 1.1-fold lower at pH 8.3, the difference increases to 1.5-fold at pH 8.7. This illustrates that the proposed approach can be used to assess the environmental risks of Ni to freshwater bodies with pH above 8.2. We believe that this 2-step procedure and the “chronic Ni bioavailability and normalization tool” is relatively easy to use and results in realistic normalized HC5 values that are also unbiased (i.e., neither over nor under-conservative).

Supplemental Data—The Supplemental Data are available on the Wiley Online Library at DOI: 10.1002/etc.3232.

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Data availability—Data, associated metadata, and calculation tools are available on request from the authors (chnys.nys@ugent.be).

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